THE GREAT INDIAN AI INTERNSHIP CHALLENGE

SOVEREIGN AI FOR INDIA

NammaVaani

A voice-first farming companion for Indian farmers

Speak naturally. Get practical farm help, with sources and confirmation.

THE IDEA A farmer speaks in Kannada, Telugu or English; NammaVaani turns the request into a focused workflow, checks relevant information and replies in speech.

THE PROBLEM

Farmers often need to switch between apps or websites for weather, market prices and local service requests. Typing, complex screens and uncertain answers can make simple tasks harder - especially when speaking is more natural than writing.

THE SOLUTION

NammaVaani is a responsive, voice-first web prototype. A farmer selects a supported language, records a question, reviews the transcript and receives a spoken answer. The app routes supported requests to focused workflows and asks a short follow-up when key details are missing.

WHAT FARMERS CAN DO

The prototype connects speech to practical, bounded tasks:

- Weather: fetch a forecast for the saved or specified place.
- Mandi prices: look up daily Agmarknet observations, with a CSV backup.
- Repair request: collect equipment, issue and village, then save a ticket after confirmation.
- Official letter draft: prepare a request for crop-protection guidance from an Agriculture Officer.
- Crop information: view a regional UAS Bengaluru Zone V crop basket.

GNANI AI INTEGRATION

Gnani Prisma v2.5 converts the farmer's selected-language recording into text (speech-to-text). NammaVaani applies its own phrase rules and workflow logic. Gnani Timbre v2.5 converts the resulting answer into speech (text-to-speech). A valid account key is required for live calls.

DESIGNED FOR TRUST

The prototype favours clear limits over confident guessing:

- Asks for missing details one question at a time.
- Shows source, location and retrieval time for weather and market information; market results include their observation date.
- Requests explicit consent before using saved GPS coordinates.
- Requires farmer confirmation before creating a repair ticket.
- Drafts a request for official crop guidance; it does not prescribe pesticides.

LOCAL RELEVANCE & DIFFERENTIATION

Built-in task replies are localised in Kannada, Telugu and English, with phrase matching for common regional expressions and some English-mixed speech. A farm profile stores state, district, taluk/mandal, hobli and village so services can use local context. The application layer connects Gnani speech APIs to transparent, useful farmer workflows rather than presenting speech recognition as the whole solution.

HOW IT WORKS

Farmer speaks > **Gnani Prisma (STT)** > **NammaVaani workflow + data** > **Clarify / confirm when needed** > **Gnani Timbre (TTS)** > **Spoken help or completed draft / ticket**

Prototype scope: Full built-in task workflows are Kannada, Telugu and English. Seven other speech locales are beta/require an optional translation bridge. Prisma auto-detection is not enabled by default. Mandi API access needs a data.gov.in key; otherwise the official CSV fallback is used. Market values are daily wholesale observations, not real-time or guaranteed farm-gate prices. Production deployment would need stronger multi-user security, privacy controls and expert review.